JECRC UNIVERSITY
School of Engineering
B. Tech; I Semester (Common to all)
Subject: Mathematics I (DMA024A)
Tutorial Sheet - 3

Time: 2 hrs. Max. Marks: 64

Instructions:

1. Attempt all the questions in sequence.
2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
3. Write the relevant question number before writing the answer.

CO1: To apply differential and integral calculus to notions of curvature and to improper integrals. Apart from some other applications, they will have a basic understanding of Beta and Gamma functions.

Part A (5×2=10)

A1. [CO1] Write the formula for the volume of a solid obtained by rotating the curve $y = f(x), a \leq x \leq b$, about the x-axis.

A2. [CO1] State the formula for the surface area of revolution of the curve $y = f(x), a \leq x \leq b$, about the x −axis.

A3. [CO1] Find the volume generated when the line $y = 3x$ is revolved about the x-axis from $x = 0$ to $x = 2$.

A4. [CO1] The curve $y = \sqrt{x}, 0 \leq x \leq 4$ is revolved about the y −axis. Write the definite integral that gives the volume of the resulting solid (no need to evaluate).

A5. [CO1] Write the integral expression for the surface area generated by revolving $y = e^x$, $x \in [0, 1]$ about the x-axis.

Part B (3×7=21)

B1. [CO1] Find the volume of the solid obtained by revolving the curve $y = \sqrt{4 - x^2}$ about the x −axis, where $x \in [0, 2]$.

B2. [CO1] Find the surface area generated by revolving the curve $y = log_e(x)$, $x \in [1, e]$ about the x −axis.

B3. [CO1] The region bounded by $y = x^2$ and $y = 0$ between $x = 0$ and $x = 2$ is revolved about the y-axis. Find the volume of the solid formed.

Part C (3×11=33)

C1. [CO1] The region bounded by $y = \sqrt{x}$ and $y = 0$ from $x = 0$ to $x = 4$ is revolved about the x −axis. Find both the volume and surface area of the solid generated.

C2. [CO1] Find the volume of the solid obtained by revolving the curve $y = sin\, x, 0 \leq x \leq \pi$, about the x −axis. Also, find the surface area of revolution for the same curve over the given interval.

C3. [CO1] A solid is generated by revolving the curve $y = \frac{1}{x}, x \in [1, 4]$, about the x −axis.

(a) Find the volume of the solid.

(b) Find the surface area of the solid.

Answers

Part A

A1: $V = \pi \int_a^b (f(x))^2 \; dx$ A2: $S = 2\pi \int_a^b f(x)\sqrt{1 + (f'(x))^2}\, dx,\ assuming\ f(x) \geq 0,$ A3: *24π.*

A4: $V = \pi \int_0^2 y^4 \, dy.$ A5: $S = 2\pi \int_0^1 e^x \sqrt{1 + e^{\{2x\}}}\, dx.$

Part B

B1: V ≈ 16.7552. B2: S ≈ 7.054966. B3: V ≈ 25.1327.

Part C

C1: 36.176903 C2: 14.423599; C3: S ≈ 9.417237 .